

Article

Perceived Importance and Quality Attributes of Automated Parcel Locker Services in Urban Areas

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Abstract: Recent global trends related to the increasing use of e-commerce are becoming a challenge for courier transport, especially in the last-mile process of delivering products to the final retail recipient. One delivery method is the personal collection of the parcel in an automated post box, available 24/7 for the customer. Our research method was based on a preliminary selection of the most important features of parcel lockers' service quality, which were extracted based on the analysis of the scientific literature and previous research. This analysis was carried out by conducting a survey of Polish parcel locker users that provided data coded according to the dimensions of the Kano model. Based on the total satisfaction index, the results allowed us to conclude that a dedicated application (−0.96), proper placement of the parcel in the box (−0.82), adjusting the size of the parcel to the size of the box (−0.79), the location of parcel stations (−0.74), and ensuring improvements for the disabled (−0.62) are the most important features in the process of the automatic delivery of parcels to recipients in urban areas. This paper enriches the literature on the customer service quality of self-service technologies for last-mile delivery with the use of automated parcel lockers.

Keywords: automated deliveries; city parcel lockers; customer service quality; courier transport; Kano model; parcel lockers; quality features



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1. Introduction

The market for services provided by courier operators experienced expansive growth in 2022. The main cause of the increase in the volume of goods delivered, especially in parcel shipments, is e-commerce, an indispensable part of global retail. One of the most important factors of this phenomenon is over five billion internet users worldwide. With the widespread availability of online shopping, retail e-commerce sales were estimated to exceed USD 5.7 trillion globally in 2022. The forecasts indicate its further growth by 56 percent over the next few years, reaching about USD 8.1 trillion by 2026 [1].

The increase in retail e-commerce services brings new challenges, especially in urban areas [2], for the logistic sphere on how to deliver specific goods to final consumers in the shortest possible time at reasonable costs. There are several different methods and approaches to the delivery of goods to the final retail recipient, depending on the specificity of the industry, the type of goods, and customer preferences, creating different policies for online-to-offline (O2O) retail strategies [3–5] or even integrating online and physical stores through omnichannel (OC) distribution [6]. The most popular ways of delivering goods to a retail customer are by using couriers; logistics service providers; e-commerce operators; or pickup at the store: BOPIS (Buy Online, Pick-Up In-Store) or other PUDO (Pick Up Drop Off) points. Along with the last possibility of delivery, there are several options that provide users freedom and convenience: point-to-point, point-to-door, or door-to-point. Such variety provides the opportunity to decide when and where recipients want to send, return, or collect parcels. This makes it possible to eliminate problems related to the delivery of goods to the retail customer, including those related to the collection and delivery to the point of sale of unwanted, advertised, or incorrectly delivered goods to the

final customer. Regarding the issue of the last kilometer in the field of delivering courier parcels, several methods of problem solving in terms of the possibility of their delivery have been proposed:

- Providing access to the house or its elements to the courier delivering the package, who places the package in an electronic terminal secured with a confirmation code;
- Leaving the package at the place of residence without the need to access the house, i.e., in the so-called home pickup box;
- Delivery of the package to a local agency, which, in turn, delivers the package to the customer's home;
- The use of pick-up points, i.e., places where the customer can pick up the parcel on his own (they can be divided into self-service and those where the release of the parcel requires service by staff).

In each of these cases, it is essential to ensure efficient logistics, shipment tracking, and communication with the customer to ensure satisfaction and timely delivery. It is important to provide support from local authorities and entrepreneurs in the aspects of spatial development in urbanized areas, where the availability of points (reception boxes, delivery boxes, controlled access system, collection points, and locker banks [7]) is one of the parameters of developing smart cities and last-mile delivery efficiency [8]. Moreover, automated parcel lockers (APLs), mailboxes, and automated parcel lockers are usually accessible 24/7, unless they are located, for example, in shopping malls, which have limited operating hours [9]. These automated lockers, parcel kiosks, locker boxes, self-service delivery lockers, and intelligent lockers are now categorized as self-service technologies (SSTs) for self-service collection and the return of goods purchased online [10]. For this reason, as predicted by industrial reports [11], parcel lockers have received positive feedback from both consumers and businesses [12,13], and the demand for parcel locker hubs is booming, especially in Sweden, Germany, Great Britain, France, and Belgium.

The largest US carrier offering this type of service today is the postal operator UPS, with 82 locker service locations at selected locations in various states at UPS Stores. In Europe, the Deutsche Post DHL Group introduced DHL Packstation in 2001, which now includes 11,500 locker stations working nationally for out-of-home (OOH) delivery. However, DHL Parcel operates in 28 European countries. The partnership between DHL and Cainiao is aimed at setting up a network of approximately 1200 parcel lockers across Poland, one of the fastest-growing e-commerce markets in Europe, with up to 40% of consumers preferring shipments delivered to parcel boxes. The largest network in Poland is currently InPost, with 20,000 automated parcel machines in Poland. There are almost as many of these as all ATMs in this country and four times more than all petrol stations [14]. In addition to Poland, this company operates under its name in Great Britain with over 3000 parcel lockers and Italy (around 300 devices) and plans to place 1000 machines in the city of Salzburg in Austria. Parcel lockers from this company are used in the networks of local couriers or postal companies in such countries as the United Arab Emirates, Estonia, Lithuania, Latvia, the Czech Republic, Slovakia, Slovenia, Germany, Iceland, Ireland, Colombia, Brazil, and the Netherlands. The Polish network of parcel lockers is the largest infrastructure of this type in Europe. As a result, 59% of Poles will find a parcel locker within a distance that can be covered in less than seven minutes on foot, which is in line with the 15 min city (FMC) concept derived from smart cities. In urban areas, 85% of them have the same or shorter distance to the parcel locker.

Currently, there are many types of parcel lockers available on the market, including indoor lockers, outdoor lockers (surface-mounted, mounted on building facades, built-in, fence boards, free-standing), dropbox lockers, refrigerated lockers, package room solutions, open locker networks, and others. They can be individual/collective, public/private, electronic/mechanic, or stationary/mobile [15]. The devices are available in hundreds of colors, materials, and sizes, according to the customer's preferences. It is important that solutions for mounted parcel lockers meet the guidelines of the PD CEN/TS 16819:2015 standard [16], which describes the technical features of parcel boxes for end use. This

covers technical features such as the size of parcels; ergonomics and safety; corrosion and water penetration resistance; and security of parcel delivery.

The traditional system of parcel lockers consists of 76 boxes, available in three sizes. Usually, parcel lockers are designed in such a way that 32/35 parcels of small size ($640 \times 380 \times 80$ mm), 32/33 parcels of medium size ($640/380/190$ mm), and 12 parcels of large ($640 \times 380 \times 410$ mm) size can be deposited in them, up to 25 kg per parcel. A system of automatic post office boxes used for sending and receiving parcels may, however, given its modular structure, have a different size, which should be adapted to the demand for services reported in a given area. Attention should be paid to the digitization of parcel locker services in smart cities, which enables contactless service using a special application on the user's smartphone. This turned out to be very convenient, especially during the pandemic.

Picking up a parcel at a parcel locker is convenient, especially for people who prefer flexibility and independence in terms of picking up goods. Parcel lockers are usually available the whole day, which allows for the collection of the parcel at a time convenient for the customer. The time slot of the delivery is one of the elements that was investigated as being crucial in decision making regarding online consumer behavior [17–19]. The other factors indicated are the speed of the delivery [20]; timeliness [21]; consumer preferences for delivery attributes in online retailing [22,23]; and, of course, delivery fees [24]. The price of the delivery service is one of the most important factors determining the need to develop a system of low-cost parcel machines in rural areas. Investigations on willingness to pay for different delivery options show that 70% of consumers are content with the cheapest form of home delivery, and 23% would pay more for same-day delivery [25]. Many scientists have conducted research in order to classify factors influencing online purchase intentions toward online shopping [26–28]; however, the economic practice shows that they are primarily fast delivery (59%), secure tracking (49%), secure packaging (36%), sustainable packaging (26%), flexible delivery (23%), and sustainable delivery routes (12%) [29].

Ultimately, the decision to use specialized devices for courier parcel delivery and collection should be based on a comprehensive analysis of these factors, considering the specific requirements and goals of the courier service provider. Regardless, the main factors that must be considered are those related to their advantages in relation to the quality of customer service, namely, proximity to infrastructure, ease of use, speed, price attractiveness, security, and continuity of service.

To address practical issues and fill a research gap, a customer perspective was adopted in the manuscript to explore and provide insight into consumer perceptions of parcel lockers and customer satisfaction with their service. This paper aims to answer the following main research questions: (1) what are the main features of automated parcel locker services identified in previous publications, and (2) which of the performance and service quality attributes of parcel distribution using parcel lockers are most important from the user's perspective? Using the example of Polish e-consumers, the study determines which quality features of the customer service process can be classified according to Kano model groups. The novel findings provide classifications of the must-be, one-dimensional, indifferent, and attractive features of the customer service process, enabling stakeholders to make rational decisions related to the development of automated parcel locker networks in cities. This paper also contributes to the literature by filling in the gap in our knowledge of customer satisfaction in relation to this widely applied self-service technology.

The study is organized as follows: Section 2 includes a brief scientific literature review of customer attitude and satisfaction with parcel lockers and the most important research achievements in this area. It covers 156 documents grouped into four thematic clusters. The analysis shows a lack of publications devoted to the identification and hierarchization of the quality factors of automated parcel locker services in urban areas. So far, authors have usually focused their research on a selected factor related to the parcel locker service process. The originality of this article lies not in focusing on analyzing these factors and selecting those that are the most important and, therefore, should be developed. This publication

fills this gap by revealing which attributes of parcel machine services are crucial according to users and which are unnecessary. Section 3 characterizes the materials and methods with descriptions of the research framework and Kano model. The Kano model was used in an innovative way to identify and prioritize quality factors related to the use of parcel lockers by estimating the satisfaction and dissatisfaction coefficients of each of them. Section 4 presents the research results of a Kano questionnaire that concerned twenty-one features of the Polish customer service quality process using parcel machines. The discussion is focused on the five most important attributes of parcel lockers and is presented in Section 5. It provides information on the current results in relation to the existing research, which emphasizes the importance of the work performed. Final conclusions are summarized in Section 6.

2. Scientific Literature Review

The analysis of the literature in the researched area was based on the resources of the Web of Science and Scopus databases. Searching the databases with the keywords “parcel locker(s)” allowed us to extract only 156 documents from 2014 until 2023, mainly articles (100), conference papers (44), book chapters (8), and other. The authors of the publications are mainly scientists from those countries where the technology of courier parcel distribution using automatic boxes is most popular: Italy (23), China (20), the United States (14), Poland (13), Germany (12), Australia (9), the United Kingdom (9), the Netherlands (8), and other. A co-occurrence analysis of all 1103 keywords from the database allowed us to construct and visualize bibliometric networks of 35 common keywords with the VOSviewer version 1.6.19 software tool, presented in Figure 1. The few cited publications concern review papers; most are original research articles. In cases concerning a selected geographical area of research described in this publication, the cited articles are case studies.

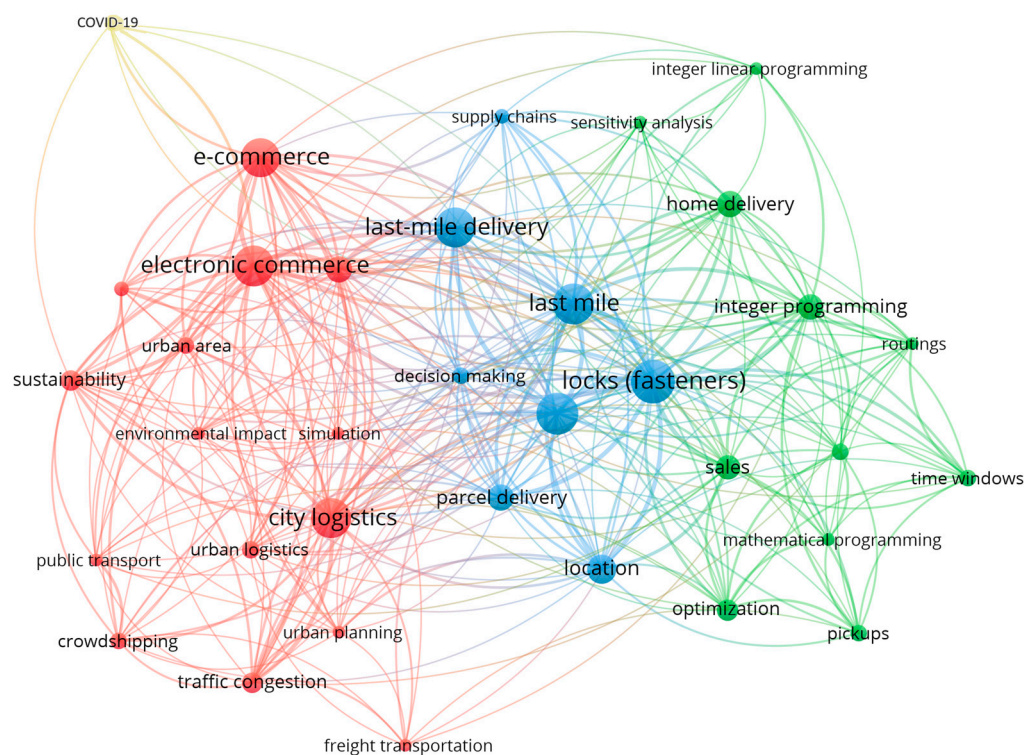


Figure 1. Bibliometric network visualization of all keywords related to parcel lockers.

The bibliometric network visualization of the keywords allowed us to identify four clusters related to parcel lockers identified with different colors. Cluster 1 (red color) refers to 15 items associated with city logistics, e-commerce, sustainability, urban area, urban plan-

ning, etc. Cluster 2 (green color), with 11 items, relates to sales, home delivery, pickups, and optimization methods (vehicle routing problems, time windows, integer linear programming, sensitivity analysis). Cluster 3 (blue color) applies to the eight items of decision making, among others: location, parcel delivery, last-mile delivery, and supply chains. The last Cluster 4 (yellow color) refers to the latest information on the recently revealed factor related to the outbreak of the COVID-19 pandemic, which effectively contributed to the development of the technology for the automatic collection of courier parcels. This analysis shows that there is a research gap in the area of customer approach regarding parcel locker users. However, some of the subject areas are identical to the qualitative characteristics perceived as satisfactory by the recipients of parcel locker services.

When analyzing the literature related to the quality of service, attention should be paid to those publications that primarily concern research on the attitude of consumers to the use of automated parcel lockers and related self-service technologies. Chen et al. [30] enhanced understanding of the components and processes that are involved in **consumer intentions** and consumer participation readiness (CPR) to use automated parcel stations (APSs).

One of the driving factors is technology anxiety, understood as the ability and willingness of customers to use self-service technologies (SSTs), as underlined by Meuter et al. [31] and Vo. et al. [32]. However, studies by Gelderman et al. [33] have shown that an even more dominant effect on the use of SSTs is the need for interaction. Moreover, as investigated by Alloulbi et al. [34], technological anxiety was found to be a significant and negative determinant of decision making in smart cities. In the results of Yuen et al. [35], the four key characteristics of innovation, relative advantage, compatibility, and trialability positively influenced customers' intention to use self-collection services. Similar results are presented in the publication by Neto and Vieira [36]. In [37], the effects of convenience, privacy security, and reliability are defined as determinants of customers' intentions to use smart lockers for last-mile deliveries. Service conveniences expressed in specific variables like network density, parking availability, spatial location, proximity to consumers' homes or offices, safe and secure operation, and hours of operation were revealed by Kedia et al. [38] and spatial accessibility was revealed by Schaefer et al. [39] as obligatory in the acceptability of new delivery technologies from a consumer perspective. Tsai and Tiwasing's [40] results revealed that convenience, reliability, privacy security, compatibility, relative advantage, complexity, perceived behavioral control, and attitude influence consumers' intentions to use smart lockers in last-mile delivery [41].

Lee and Lyu [42] examined **personal values** and **consumer traits** as antecedents of attitudes toward an intention to use SSTs from a need for interaction and self-efficacy. According to Milioti et al. [43], consumer characteristics like environmental awareness can be an effective subject of consideration in modeling consumers' acceptance of a click-and-collect service. Understanding **consumer preferences** in relation to the selected attributes of deliveries, such as location, delivery time, information and traceability, cost of transportation, and the willingness to use automatic delivery stations are, according to [44], factors necessary to estimate the demand for new technologies in urban areas. According to Rai et al. [45], APS solutions are also growing in popularity because consumers choose them because of previous disappointment resulting from failed delivery problems observed in the urban distribution of goods.

The above research on the key factors related to consumer intentions, traits, and preferences influencing the choice of APS technology was carried out through a statistical analysis of survey studies in various parts of the world. However, Lai et al. [46] introduced the service quality (SERVQUAL) model and the logistics service quality (LSQ) model; this study investigates the antecedents of customer satisfaction with parcel locker services in last-mile logistics. The analysis of the scientific literature led to the conclusion that there were no earlier publications using the Kano model for analysis in the selected area.

Understanding consumer attitudes influencing the choice of APS technology in last-mile delivery has a direct and indirect impact on customer satisfaction with the service

provided. However, customer loyalty, customer retention, and profitability are elements of firm operations, which, in turn, are related to service quality as defined by Gupta [47] and Li and Shang [48]. Although extant studies have argued that consumer intentions and preferences clearly impact parcel lockers as self-collection services, scant evidence has been provided on the influencing factors of customer satisfaction regarding parcel locker service quality. However, some of the features can be distinguished in the previous literature. Thus, the scientific objective of this study is to identify and refine the quality attributes of automated parcel locker services in urban areas to fill the gap in knowledge in this area.

3. Materials and Methods

3.1. Research Framework

The research part of this paper is focused on the analysis of APS services in urban areas from the point of view of their users. The main objective of the study was to gain information on the hierarchy of customer expectations toward parcel machines, which may be helpful in improving services in the field of deliveries with their use and also in spatial planning or making decisions regarding the development of this type of infrastructure in cities. The opinions of users of parcel locker services were examined using a questionnaire using the diagnostic survey method.

The research planning included four stages, presented with a framework in Figure 2.

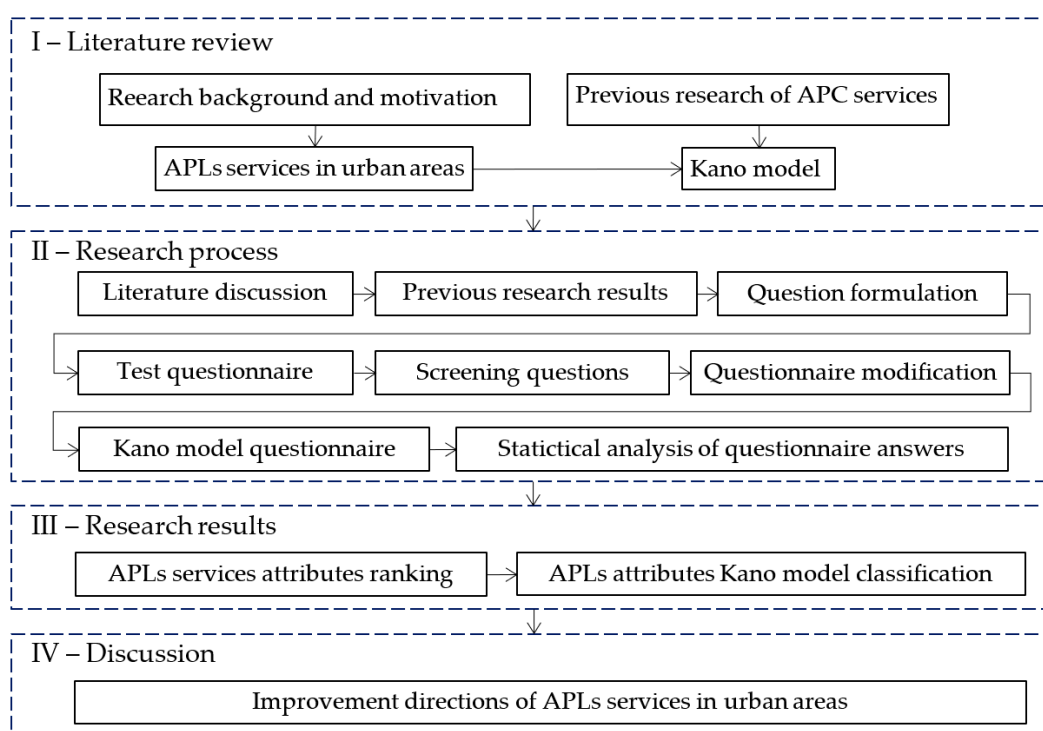


Figure 2. Research framework for an analysis of automated parcel locker service attributes.

Stage I was the initial stage, focused on review. As a result, the assumptions and objectives of automated parcel locker service attribute analysis were developed. This part was implemented in two tracks. The first part was the scientific literature review, the most important research achievements of customer attitude and satisfaction with parcel lockers, and the methods used to analyze them. In the second part, the achievements of previous analyses were used. The joint results of this part of the methodology enabled the decision to use the Kano model, which has never been used before to classify APL service attributes.

Stage II was focused on the research process, wherein the procedure was carried out according to the Kano model. To obtain answers, we decided to use the survey method. It allowed us to collect data cost-effectively and reliably because of the anonymity.

Based on the knowledge related to the literature review and previous university research results [49–53], preliminary questions were developed for the survey based on the needs and expectations of the users of parcel machine systems in cities. In the next part of the research, a questionnaire test was carried out, which allowed us to check the respondents' understanding of the questions. After the tests, we decided to modify the survey layout for a more transparent reception of questions. Then, a survey was carried out in accordance with the Kano model, followed by a statistical analysis of the obtained results. Conducting the Kano questionnaire concerned twenty-one features of the customer service quality process using parcel machines. Given the automation of the process of collecting and sending parcels and the use of a dedicated mobile application, an online survey with Google Forms was also used for this research. This allowed the survey questionnaire to reach people who consciously use e-commerce and freely navigate the internet. The survey form in question consisted of 24 closed, single-choice questions. The survey was conducted at the turn of November and December 2022. In total, 468 respondents from the Silesian Metropolis (the southern part of Poland) took part in the survey, of which 55.6% were women, which corresponds to the total number of 260 people surveyed. Out of the 468 people participating in the study, as much as 65% (304 people) were people aged 18–24. The second largest group in terms of share, constituting 23.1%, were people aged 25–34 (108 people). Another 3.4% of all respondents were in the age range of 35–44 (16 people), while 4.3% were in the age range of 45–54 (20 people). Twenty people under the age of 18 corresponded to 4.3% of all respondents.

To calculate the sample size of the population for the determination of the minimum sample size, which is required in order to know the adequate or correct proportion of the population, along with the confidence level and the margin of error, the following formula, based on [54], was used:

$$n = N \cdot \frac{\frac{Z^2 \cdot p \cdot (1-p)}{e^2}}{\left[N - 1 + \frac{Z^2 \cdot p \cdot (1-p)}{e^2} \right]}, \quad (1)$$

where n —sample size, N —population size, Z —critical value of the normal distribution at the required confidence level, p —sample proportion, e —margin of error.

Determination of the sample size necessary to estimate the proportion of people using parcel lockers in the Silesian Metropolis (with 2,249,568 residents in 2022) with 95% confidence was assumed with a margin of error of 5% and a population proportion of 0.5. Based on the formula above, the sample size estimated was 384. This allowed us to conclude that the survey results were representative of the number of responses obtained in the research. The survey was conducted based on the principle of simple random sampling, which enables the selection of people randomly, all with the same probability. Before starting the actual study using the Kano questionnaire, the survey data included information related to gender, age, place of residence, and education. The sample was considered representative of the population from which it was drawn because the aggregate characteristics of the sample closely reflected the same aggregate characteristics of the population.

Stage III of the methodology was concentrated on result analysis. According to the questionnaire survey results, a Kano model was established to obtain the satisfaction ranking of automated parcel locker service attributes.

Finally, stage IV covered the discussion of the results. Further conclusions related to the development and possible improvements of APLs in urban areas were made.

3.2. Kano Model

Given its popularity, accessibility, and transparency, the Kano model was chosen as the optimal method for the analysis and classification of the most important attributes of courier service quality, including the delivery of parcels through automated sending-and-receiving parcel lockers. It is based on posing logical and unambiguous questions that determine the group for which a given distribution process is dedicated. The Kano

model was chosen as one of the tools that allowed us to find out what features of parcel lockers need immediate attention to improve service realization based on performance enhancement and customer satisfaction levels. It allows us to understand the customer's emotional responses to service loyalty and its features. The classification and prioritization of attributes enable an increase in customer satisfaction with features that customers desire.

The Kano model is a two-dimensional modeling tool (satisfaction/sufficiency—see Figure 3) proposed by Noriaki Kano of the Tokyo Institute of Technology [55], mainly used for user requirement classification and prioritization classes. The use of the Kano system is practically reflected in the modeling of many processes, including services and objects. The analysis was supported with a review of papers on the most commonly used approaches to the classification of quality attributes according to the Kano models published by Berger et al. [56], Wang and Ji [57], Mikulić and Prebežac [58], Wittel et al. [59], and Kermanshachi et al. [60].

The Kano model was previously successfully used as a technique for classifying quality attributes in a whole range of research areas, like hospitality services by Zobnina and Rozkhov [61], the healthcare industry by Gupta and Srivastava [62], restaurant services by Pai et al. [63] and Chen et al. [64], elderly care service platforms by Zhou et al. [65], and even hotel service robots by Xie et al. [66,67].

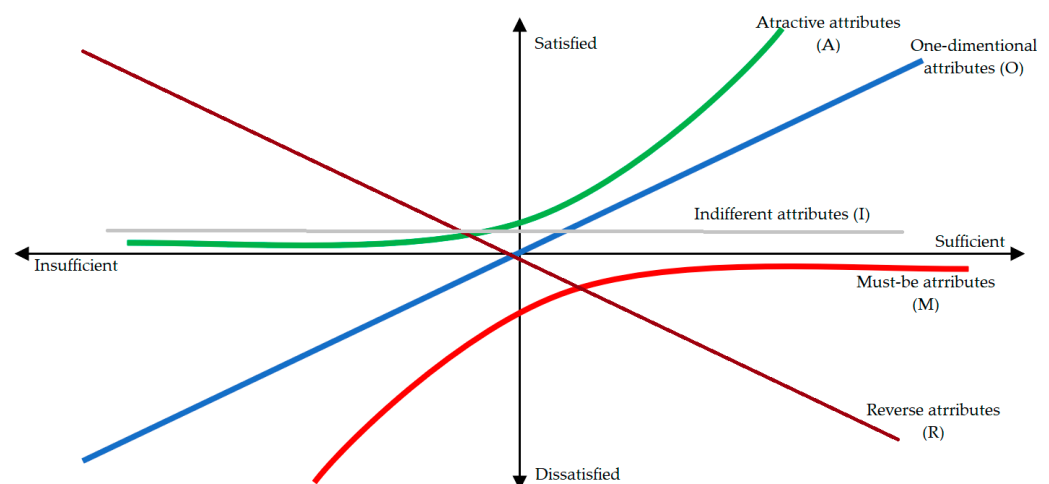


Figure 3. Attribute groups of a two-dimensional Kano model. Source: own elaboration based on [55,65,68].

The main assumption of the Kano model is that not all attributes of the object or service are equally important to the customer. Three basic groups of features should be distinguished in the Kano model:

- **Must-be attributes (M).** This is a set of requirements that the recipient is not aware of, but they are extremely important from the perspective of shaping his satisfaction or dissatisfaction. This group is characterized by requirements whose fulfillment will not increase customer satisfaction, but their absence will result in customer dissatisfaction;
- **One-dimensional attributes (O).** This is a group of desired customer requirements. This means that their implementation increases satisfaction, but failure to meet these requirements will result in dissatisfaction;
- **Attractive attributes (A).** These are the service requirements meant to attract attention. Their fulfillment has a huge effect on increases in satisfaction, while their failure will affect the customer's feelings.

In addition to the above-mentioned basic groups of features, Kano also defined the so-called additional requirements, which are more difficult to identify given their specificity. These include the following:

- Indifferent attributes (*I*). The presence or absence of these requirements will in no way affect a return or reduction in customer satisfaction. They are irrelevant;
- Questionable attributes (*Q*). This is a group of requirements about which there is no reliable information on whether they are relevant to the consumer;
- Reverse attributes (*R*). These appear when the opposite of a given feature is important to the client.

According to the refined theory of the Kano model presented by Lo [68], a product or service pays more attention to whether the must-be quality, one-dimensional quality, and attractive quality are satisfied or not. Various classification methods were previously proposed with a cross-reference of consumer options. Schvaneveldt et al. [69], as well as Matzler and Hinterhuber [70], proposed the five-grade evaluation. The last, verified methodological approach was adapted to this study. According to the Kano model, each question is asked in two variants for the examined attitude, as presented in Table 1. In the positive variant, a question is built describing that something occurs, works, and is sufficient. On the other hand, in the negative variant, the problem is that something is absent, does not work, or is inefficient.

Table 1. Variants of the questionnaire presenting the method of building questions.

Tested attitude	Positive variant	(1)	I like it that way
		(2)	It must be that way
		(3)	I am neutral
		(4)	I can live with it that way
		(5)	I dislike it that way
	Negative variant	(1)	I like it that way
		(2)	It must be that way
		(3)	I am neutral
		(4)	I can live with it that way
		(5)	I dislike it that way

Based on the number of responses received within one feature, individual categories (*A*, *M*, *O*, *R*, *Q*, or *I*) are determined, as presented in Table 2. It is important to eliminate potential interference resulting from incorrectly formulated questions and their poor understanding by respondents. To do this, the following two rules must be followed:

- If $(A + M + O) > (R + Q + I)$, then, finally, the category from groups *A*, *M*, or *O* with the highest value is selected;
- If $(A + M + O) < (R + Q + I)$, then, finally, the category from the *R*, *Q*, or *I* group with the highest value is selected.

Table 2. Evaluation and interpretation of questionnaire results.

Tested Attitude		Negative Variant				
		(1) I Like It That Way	(2) It Must Be That Way	(3) I Am Neutral	(4) I Can Live with It That Way	(5) I Dislike It That Way
Positive variant	(1) I like it that way	<i>Q</i>	<i>A</i>	<i>A</i>	<i>A</i>	<i>O</i>
	(2) It must be that way	<i>R</i>	<i>I</i>	<i>I</i>	<i>I</i>	<i>M</i>
	(3) I am neutral	<i>R</i>	<i>I</i>	<i>I</i>	<i>I</i>	<i>M</i>
	(4) I can live with it that way	<i>R</i>	<i>I</i>	<i>I</i>	<i>I</i>	<i>M</i>
	(5) I dislike it that way	<i>R</i>	<i>R</i>	<i>R</i>	<i>R</i>	<i>Q</i>

A: attractive; I: indifferent; M: must-be; O: one-dimensional; Q: questionable; R: reverse.

To better understand the attitude of APL users, it is important to acknowledge all the responses when evaluating and categorizing each factor. For this reason, Shahin et al. [71] revised the satisfaction coefficient (C_{Si}) and dissatisfaction coefficient (C_{Di}) used in the Kano model and advised using the following ones:

$$C_{Si} = (A + O) / (A + O + M + I), \quad (2)$$

$$C_{Di} = (O + M) / (A + O + M + I)(-1), \quad (3)$$

A larger satisfaction coefficient indicates higher consumer satisfaction, while a smaller dissatisfaction coefficient means higher dissatisfaction. The difference between the values of the satisfaction and dissatisfaction coefficients is known as the total satisfaction index, and the attributes can be ranked based on that calculated total value.

4. Results

The Kano questionnaire concerned the following twenty-one features of the customer service quality process using parcel machines:

- 24/7 customer support;
- Additional parcel locker services (e.g., refrigerated lockers, laundry services);
- Adjusting the size of the package to the size of the box (parameter related to problems with removing the parcel when the parcel tightly fills the box);
- Advertisements on parcel lockers;
- The convenience of receiving and sending parcels;
- Dedicated applications;
- The ease of use of the parcel locker;
- Improvements for people with disabilities;
- Methods of parcel pick-up (standard or automatically via smartphone);
- Methods of parcel drop-off (standard or online without the need to print the shipping label, which is attached by the courier upon receipt);
- Natural environment awareness;
- Parcel locker locations (related not only to a satisfactory distance but also to the place, for example, close to work or school);
- Parcel locker novelty (parameter related to modernity because newly created APLs sometimes differ from previous ones);
- Parcel locker service time (time needed for parcel pick-up or drop-off);
- Parking next to the parcel locker;
- Placing the parcel in a specific box (the service is called the easy access zone and enables users to place parcels in the lower compartments in the parcel locker);
- The possibility of using a multi-locker (allows users to pick up multiple parcels from multiple senders by one receiver or drop off multiple parcels to one receiver);
- Security of the parcel placed in the locker;
- Size of the parcel locker (number of boxes);
- Temperature inside;
- Time window for parcel pick-up.

Table 3 presents the proportion of questionnaire answers on each type of quality attribute in the Kano groups and then the classification decisions.

Based on the results from the respondents related to the Kano questionnaire, it was possible to classify five attributes into the must-be (M) type, four attributes into the one-dimensional (O) type, three attributes into the attractive (A) type, and nine attributes into the indifferent (I) type. When analyzing the percentage distribution of responses, no attitudes were assigned to the questionable (Q) or reverse (R) types.

Table 3. Kano model classification of automated parcel locker service attributes.

Attribute	Proportion of Each Type of Kano Quality Attribute (%)						Classification Type
	<i>M</i>	<i>O</i>	<i>A</i>	<i>I</i>	<i>Q</i>	<i>R</i>	
24/7 customer support	20.52	5.13	14.53	57.25	0.86	1.71	<i>I</i>
Additional APL services	11.11	35.89	6.87	10.26	3.42	32.48	<i>O</i>
Adjusting the size of the package to the size of the box	54.70	7.69	8.55	28.21	0.00	0.85	<i>M</i>
Advertisements on APLs	19.66	15.38	12.82	35.04	1.71	15.38	<i>I</i>
Convenience of receiving and sending parcels	7.69	5.13	61.54	11.97	2.56	11.11	<i>A</i>
Dedicated applications	57.26	14.53	8.55	18.80	0.00	0.85	<i>M</i>
Ease of use of APLs	19.66	31.62	12.82	17.95	0.00	17.95	<i>O</i>
Improvements for people with disabilities	30.77	7.69	15.38	45.30	0.00	0.85	<i>M</i>
Methods of parcel pick-up	24.79	10.26	27.35	35.04	0.85	1.71	<i>A</i>
Methods of parcel drop-off	7.69	3.42	20.51	59.83	6.84	1.71	<i>I</i>
Natural environment awareness	19.66	5.98	16.24	57.26	0.00	0.85	<i>I</i>
Parcel locker location	42.74	6.84	13.68	31.62	2.56	2.56	<i>M</i>
Parcel locker novelty	15.38	19.66	7.69	52.14	0.85	4.27	<i>I</i>
Parcel locker service time	21.37	16.24	36.75	21.37	1.71	2.56	<i>A</i>
Parking next to APL	18.80	52.14	6.84	20.51	0.00	1.71	<i>O</i>
Placing the parcel in a specific box	43.59	12.82	10.26	29.91	0.85	2.56	<i>M</i>
Possibility of using a multi-locker	20.51	11.11	14.53	36.75	1.71	15.38	<i>I</i>
Security of the parcel placed in the APL	14.53	43.59	17.95	13.68	2.56	7.69	<i>O</i>
Size of the APL	11.11	6.84	19.66	58.12	0.00	4.27	<i>I</i>
Temperature inside the APL	19.66	4.27	10.26	59.83	3.42	2.56	<i>I</i>
Time window for parcel pick-up	5.98	1.71	11.11	64.96	2.56	13.68	<i>I</i>

Must-be attributes are the ones that must be provided for APL users. Service providers need to regard this quality pattern when developing delivery services with parcel locker usage. Dedicated application and the possibility of placing the parcel in a specific box for people with disabilities are extremely important issues for consumers. Furthermore, analyzing the locations of parcel locker stations in the city is crucial. Training couriers in the selection of a specific box in a parcel locker when placing a parcel has a significant impact, according to the respondents, on the ease of pick-up and, thus, on the quality of the service.

The one-dimensional attributes indicate the adequacy of quality for consumer satisfaction. They are used as a basis for APL service providers to differentiate prices. The customers consider the basic functionality of parcel lockers to be important, which suggests that operators must maintain and improve the one-dimensional quality attributes of additional services offered, thus ensuring parcel safety or providing parking places next to automated parcel stations.

The attractive attribute is used as a tool to understand service differentiation and reflects a consumer's desire for a specific feature. The results of this study revealed that consumers were not satisfied with but accepted features related to operation time, convenience of use, and the possibility of parcel pick-ups with different methods.

The indifferent attributes have little effect on customer satisfaction. According to the research results, this was the biggest group of features related to advertisements put on APLs, the possibility of using a multi-locker, 24/7 customer support, the temperature inside, or the time window for parcel pick-up.

The consumer satisfaction coefficient (C_{Si}) and dissatisfaction coefficient (C_{Di}) were used to evaluate the degree of customer satisfaction and dissatisfaction with the APL service attributes. The total satisfaction calculation allowed us to present the ranking of attributes according to their classification types in Table 4. As in Xu et al. [72], the matrix presented in Figure 4 illustrates these relationships.

Table 4. Ranking of automated parcel locker service attributes based on total satisfaction index.

Classification Type	Attribute	Satisfaction Coefficient (C_{Si})	Dissatisfaction Coefficient (C_{Di})	Total Satisfaction Index	Ranking
Must-be (M)	Dedicated application	0.2328	−0.7241	−0.9569	1
	Placing the parcel in a specific box	0.2389	−0.5841	−0.8230	2
	Adjusting the size of the package to the size of the box	0.1638	−0.6293	−0.7931	3
	Parcel locker location	0.2162	−0.5225	−0.7387	4
	Improvements for people with disabilities	0.2328	−0.3879	−0.6207	5
One-dimensional (O)	Additional APL services	0.6667	−0.7333	−1.4000	1
	Security of the parcel placed in APL	0.6857	−0.6476	−1.3333	2
	Parking next to APL	0.6000	−0.7217	−1.3217	3
	Ease of use of APLs	0.5417	−0.6250	−1.1667	4
Attractive (A)	Parcel locker service time	0.5536	−0.3929	0.9464	1
	Convenience of receiving and sending parcels	0.7723	−0.1485	0.9208	2
	Methods of the parcel pick-up	0.3860	−0.3596	0.7456	3
Indifferent (I)	Advertisements on APLs	0.3402	−0.4227	−0.7629	1
	Possibility of using a multi-locker	0.3093	−0.3814	−0.6907	2
	Parcel locker novelty	0.2883	−0.3694	−0.6577	3
	Natural environment awareness	0.2241	−0.2586	−0.4828	4
	24/7 customer support	0.2018	−0.2632	−0.4649	5
	Size of the APL	0.2768	−0.1875	−0.4643	6
	Temperature inside	0.1545	−0.2545	−0.4091	7
	Methods of parcel drop-off	0.2617	−0.1215	−0.3832	8
	Time window for parcel pick-up	0.1531	−0.0918	−0.2449	9

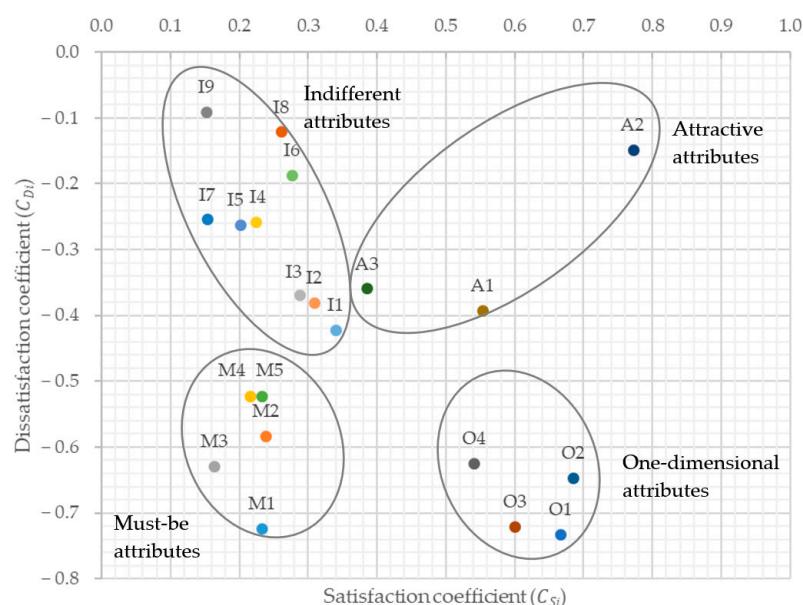


Figure 4. Satisfaction/dissatisfaction coefficient matrix with groups of attributes in the Kano model.

5. Discussion

The research carried out using the Kano model allowed us to obtain answers to the research questions related to the attributes of parcel locker services important to users and enabled their classification. The research conducted allowed us to not only understand the perception of individual features by users but also to rank them. The greatest merit of the research was the identification of five out of twenty-one attributes of parcel lockers, which, according to the Kano model, were considered the most important.

The most important factor necessary to maintaining the high quality of services related to parcel lockers is a **dedicated application**. The most common in Poland, the InPost Mobile application, was created to improve convenience for customers and expand the possibilities of managing their shipments. The application is free and available for download in the Google Play Store, the Apple App Store, and the Huawei AppGallery. Since its introduction in 2019, the application has gained several improvements and useful features, such as the possibility of parcel tracking, remote opening the parcel box (this service was the preferred form of service, especially during the COVID-19 pandemic), the possibility of shipping a parcel without a printed label, notifications about packages ready for pick-up, the possibility of extending the pick-up time, easy access zones, dynamic redirection, quick returns, sharing parcel information between application users, cashless payments for cash on delivery (COD) parcels, maps of InPost APLs, and navigation to the specified parcel locker. Mobile applications are indispensable because, as Tang et al. [73] emphasize, the adoption of Internet of Things (IoT)-based smart parcel lockers has developed very rapidly. Niederprüm and van Lienden [74] underline that operators have found innovative solutions for last-mile delivery in order to improve access and operability, including the use of advanced technologies through the use of mobile applications, e.g., integrating electronic systems and the wireless transfer of information between online retailers, deliverers, and recipients, as well as enabling contactless parcel pick-up. Similarly, Akdoğan and Özceylan [75] note that parcel locker applications can meet some of the alternative needs of environmentally friendly and efficient solutions for logistics services in harmony with the digitalizing world in order to enable **ease of use of APLs** (the one-dimensional attribute).

The next must-be attributes of parcel locker services are of a technical nature. The feature connected to **adjusting the size of the package to the size of the box** is certainly related to the negative experiences of recipients with their parcels getting stuck in a locker or problems with their removal. Parcel locker stations (terminals) are constructed in different sizes in terms of the number of lockers they contain, ranging from compact terminals consisting of less than 15 lockers to large terminals that may contain more than 500 lockers (the largest parcel locker station in Europe is in Finland with 1002 lockers [74]). Parcel locker modules, adapted to parcels of various sizes, are configured to contain compartments of various sizes, adjusted to three standard dimensions: small, medium, and large compartments. Typical locker dimensions can come in ranges of 8–75 cm × 35–40 cm × 60–64 cm depending on the supplier, and the configuration of the modules can generally be customized to include a certain number of specific cabinet sizes within the overall dimensions of the module. This is one of the factors influencing the multi-objective green express cabinet assignment problem in urban last-mile logistics presented by Ji et al. [76]. Incorrectly matching the dimensions can lead to several problems, because the package will not fit into a too small locker. Parcel lockers are adapted to individual shipments, which is why it is important to provide clear information to users to avoid later problems. It is also good to provide users with brief instructions about what to do when the chosen parcel locker size is too small for a drop-off service. Stojanov [77] emphasizes that one of the limitations of parcel lockers is that not all parcels fit within the specified dimensions of each APL and, therefore, will require alternative delivery methods.

The issue of **placing the parcel in a specific box** relates to the easy access zone service, which allows for the delivery of parcels to lower-placed lockers at a height between 30 cm and 150 cm. This attribute of service is dedicated to users of short stature or for people with physical disabilities. According to the research results, **improvements for people with**

disabilities or special needs must also be considered by parcel locker operators. As noted by Lagorio and Pinto [78], any inconvenience or barriers arising in the service process can be easily overcome by providing telephone customer service connected to the lockers and the inclusion of voice/commands for people with disabilities.

In view of the emergence of various operators operating parcel machines, it is worth looking at the one-dimensional attributes of their services, which were indicated by the respondents, because they can contribute to gaining a competitive advantage. The first reported factor is **additional APL services**. Such ideas concerned, for example, the introduction of the Hi'Shine clothes dry cleaning service in InPost parcel machines, but they were withdrawn from the market. Another is to enable the delivery of shipments at a controlled temperature. Wróbel-Jędrzewska and Polak [79] propose the construction of a prototype device (food parcel locker) consisting of small cooling and freezing boxes ensuring temperature conditions (+5 °C or −18 °C) to enable grocery storage.

Another attribute, the **security of the parcel placed in APL**, is widely discussed by researchers. For example, the study by An et al. [80] examined consumer decisions to select parcel locker services regarding technology assessment and privacy protection concerns. The work of Min-Hye et al. [81] suggests a double-security smart parcel locker to keep parcels safe from the risk of loss and personal information leakage by using parcel invoice barcodes through near-field communication (NFC) modules for smartphones. However, most recipients, as stated by Keen et al. [82], perceive the level of security of parcel locker services as satisfactory.

The next suggestion, **parking next to APL**, is related to the widely analyzed issue of the location of parcel lockers. When choosing locker installation places, Guerrero-Lorente et al. [83] suggest that high-demand points should be considered to obtain higher efficiency, such as subway or bus stops, shopping centers, supermarkets, etc. Iyer et al. [84] noticed that, for high-density urban areas, the optimal walking distance can vary widely across the network, and this solution can deliver varying levels of benefits depending on the location of parcel lockers. For this reason, the possibility of parking at the parcel locker, according to Lachapelle et al. [85], may be a decisive variable for this form of parcel delivery. As demonstrated by Wang et al. [86], movable locker units with few lockers can be more suitable for scattered low-demand areas. Furthermore, the approach of Schwerdfeger and Boysen [87], as well as Lazarević et al. [88], proved that mobile parcel lockers (MPLs) allow for easy access to hard-to-reach destinations and, especially when driving autonomously, reduce customer inconvenience connected with walking or car parking.

The results of this study on indifferent attributes, which have little effect on customer satisfaction, are surprising. It turns out that many of them are the subject of research by scientists and may in the future become factors of greater importance for users of parcel machines.

6. Conclusions

The main reason for the increase in the volume of shipments recorded by courier operators, visible especially in parcel shipments, is global electronic commerce, creating new trends in delivering goods purchased online to customers. An increase in retail e-commerce services brings new challenges, especially in urban areas, for the logistic sphere of different delivery possibilities in the shortest possible time and at reasonable costs. In view of new organizational and infrastructural solutions regarding the use of automated parcel machines in the distribution of goods, the issue of the quality of services and the needs of users becomes important.

To fill a knowledge gap, a customer perspective was adopted in this paper to explore and provide insight into consumer perceptions of parcel lockers and features of customer satisfaction with their services. Thus, the scientific objective of this study was to identify and refine the quality attributes of automated parcel locker services in urban areas. A methodology based on the Kano model was adopted for the analysis, which has proven itself as a research method in studies on different subjects. The questionnaire, which is a

research tool, included twenty-one attributes related to the quality of parcel locker services and was conducted on 468 respondents.

Based on the Kano questionnaire results, it was possible to classify five attributes into the must-be (M) type, four attributes into the one-dimensional (O) type, three attributes into the attractive (A) type, and nine attributes into the indifferent (I) type. It was also possible to classify the rankings of attributes into groups based on satisfaction and dissatisfaction coefficients. The must-be attributes, which should become mandatory for service providers, are, according to users, parcel stations, ensuring improvements for the disabled, adjusting the size of the parcel to the size of the box, the proper placement of the parcel in the box, and a properly functioning dedicated application. It is, therefore, recommended that these factors be considered by stakeholders in the development of parcel locker services in cities. Additional resources should be devoted to research on the development of the functionalities of mobile applications, facilitating the operation of parcel lockers. The suggested solution is a series of ongoing training sessions for the couriers responsible for fitting parcels into parcel lockers properly, thus avoiding problems for the recipient when removing a parcel. Stakeholders responsible for issuing permits for the development of infrastructure related to parcel lockers should pay special attention to their location and ensure architectural accessibility for people with disabilities.

It is also worth paying attention to one-dimensional attributes that can attract users, such as additional services, the security issues of parcels in lockers, parking zones provided next to parcel stations, and the ease of use of APLs. The other attractive and indifferent attributes should not be underestimated because they become a very important carrier of information for decision-makers related to the spatial development of cities and entrepreneurs providing services related to the distribution of parcels.

A limitation of this study is that it is based on a convenience sample. Although a large sample of 468 respondents was used, and all interest groups were well represented, older people were underrepresented in the research sample. While the possibility cannot be ruled out that this underrepresentation may have influenced the results to some extent, the fact is that, when shopping online, young users dominate statistics. Therefore, it may be advisable to repeat this study on a larger and more representative sample. Repeating the research in this direction would also be rational given that, with the passage of time, other factors of the environment may change, influencing the decision making of parcel machine users and, in particular, the perception of quality attributes. Interesting results may also be presented when analyzing customer attitudes in cities in different countries, where their decisions are influenced by other cultural, religious, belief, and other factors.

Further research work on the subject matter may concern the development of a decision-making model covering subsequent areas of parcel locker service quality attributes. An interesting extension would be the analysis and verification of the comprehensiveness of these services, the range and flexibility of which are very diverse.

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Abbreviations

APLs	Automated parcel lockers
APS	Automated parcel stations
BOPIL [®]	Buy Online, Pick-up in Locker
BOPIS	Buy Online, Pick-Up In-Store
COD	Cash on delivery

CPR	Consumer participation readiness
FMC	15 min city
IoT	Internet of Things
MPL	Mobile parcel locker
NFC	Near-field communication
O2O	Online-to-offline
OC	Omnichannel
OOH	Out-of-home
PUDO	Pick Up Drop Off
SSTs	Self-service technologies

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